MATHS MANTHAN
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 10 (QUADRATIC EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 13(iii).
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions (VSAQ) of 2 marks each.
4. **Section C** comprises 2 Short Answer questions (SA) of 3 marks each.
5. **Section D** comprises 2 Long Answer questions (LA) of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A (Multiple Choice & Assertion-Reason Questions 6 × 1 = 6 Marks)

Q1. If the product of two consecutive positive integers is 306, then the integers are: [1]
(A) $16, 17$ (B) $17, 18$ (C) $18, 19$ (D) $15, 16$

Q2. The discriminant of the quadratic equation $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$ is: [1]
(A) 32 (B) -32 (C) 16 (D) 24

Q3. The roots of the quadratic equation $2x^2 - 5x + 3 = 0$ are: [1]
(A) $1, \frac{3}{2}$ (B) $-1, -\frac{3}{2}$ (C) $2, 3$ (D) $1, \frac{2}{3}$

Q4. If p and q are the roots of the equation $x^2 + px + q = 0$, then: [1]
(A) $p = 1,\ q = -2$ (B) $p = 0,\ q = 1$ (C) $p = -2,\ q = 1$ (D) $p = -2,\ q = 0$

Q5. If one root of the equation $ax^2 + bx + c = 0$ is three times the other, then $b^2 : ac =$: [1]
(A) 3 : 1 (B) 3 : 16 (C) 16 : 3 (D) 16 : 1

Q6. Assertion-Reason: [1]
Directions: (A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.
Assertion (A): If the coefficient of x^2 and the constant term have opposite signs, the equation $ax^2 + bx + c = 0$ always has real roots.
Reason (R): When a and c have opposite signs, $ac < 0 \implies -4ac > 0$, making $D = b^2 - 4ac > 0$ strictly positive.

SECTION B (Very Short Answer Questions 2 × 2 = 4 Marks)

Q7. Determine whether the quadratic equation $2x^2 + 5\sqrt{3}x + 6 = 0$ has real roots, and if so, find them using the quadratic formula. [2]

Q8. Find the values of k for which the quadratic equation $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots. [2]

SECTION C (Short Answer Questions $2 \times 3 = 6$ Marks)

Q9. Solve the following equation for x: [3]

$$\frac{x+1}{x-1} + \frac{x-2}{x+2} = 3, \quad x \neq 1, -2$$

Q10. A motorboat whose speed in still water is 18 km/h takes 1 hour more to travel 24 km upstream than to return downstream to the same spot. Determine the speed of the stream. [3]

SECTION D (Long Answer / HOTS Questions $2 \times 5 = 10$ Marks)

Q11. A fast express train takes 3 hours less than a slow passenger train for a journey of 600 km. If the speed of the slow train is 10 km/h less than that of the fast train, formulate the quadratic equation and determine the average speeds of both trains. [5]

Q12. If the roots of the quadratic equation $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are equal, prove that: [5]

$$a = b = c$$

SECTION E (Applied Case Study Problem $1 \times 4 = 4$ Marks)

Q13. Case Study: Urban Parks Landscaping & Triangular Garden Pavilion

A municipal botanical park is constructing a triangular floral pavilion in the shape of a right-angled triangle. Environmental landscaping regulations specify that the horizontal base of the triangular pavilion must be 2 metres longer than twice its vertical altitude h metres. The total ground area enclosed by the floral pavilion is designed to be exactly 30 m^2.

(i) Express the horizontal base of the triangular pavilion in terms of the vertical altitude h. [1]

(ii) Set up the quadratic equation in h representing the total enclosed area of the pavilion. [1]

(iii) Solve the equation to determine the altitude h, the base, and the hypotenuse of the pavilion. [2]

OR

If ornamental perimeter kerbing costs Rs. 120 per metre, calculate the total cost of installing the decorative kerbing around all three sides of the triangular pavilion.

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